

Blood Relations

A Comprehensive Guide to Concepts, Family Tree Diagrams, and Problem Solving Strategies



RAMAGIRI MANISH

Roll No:

25MVCAIR0069

MODULE 01

Fundamental Concepts

Understanding family ties, maternal vs. paternal sides, and generational hierarchy

Relationship Types



Paternal Relations

Relationships established through the father's side of the family. Includes paternal grandfather, paternal grandmother, paternal uncle (father's brother), and paternal aunt.



Maternal Relations

Relationships connected through the mother's side. Includes maternal grandfather, maternal grandmother, maternal uncle (mother's brother), and maternal aunt.



Relations by Marriage

Relationships created via marital alliance, denoted by the suffix '-in-law' (e.g., father-in-law, sister-in-law, son-in-law, brother-in-law).

Generational Hierarchy

Upper Generations (+1 & +2)

Generation +2: Grandfather, Grandmother (Paternal & Maternal).

Generation +1: Father, Mother, Uncle, Aunt, Father-in-law, Mother-in-law.

Upper generations are placed at higher levels in standard family trees.

Lower Generations (0, -1 & -2)

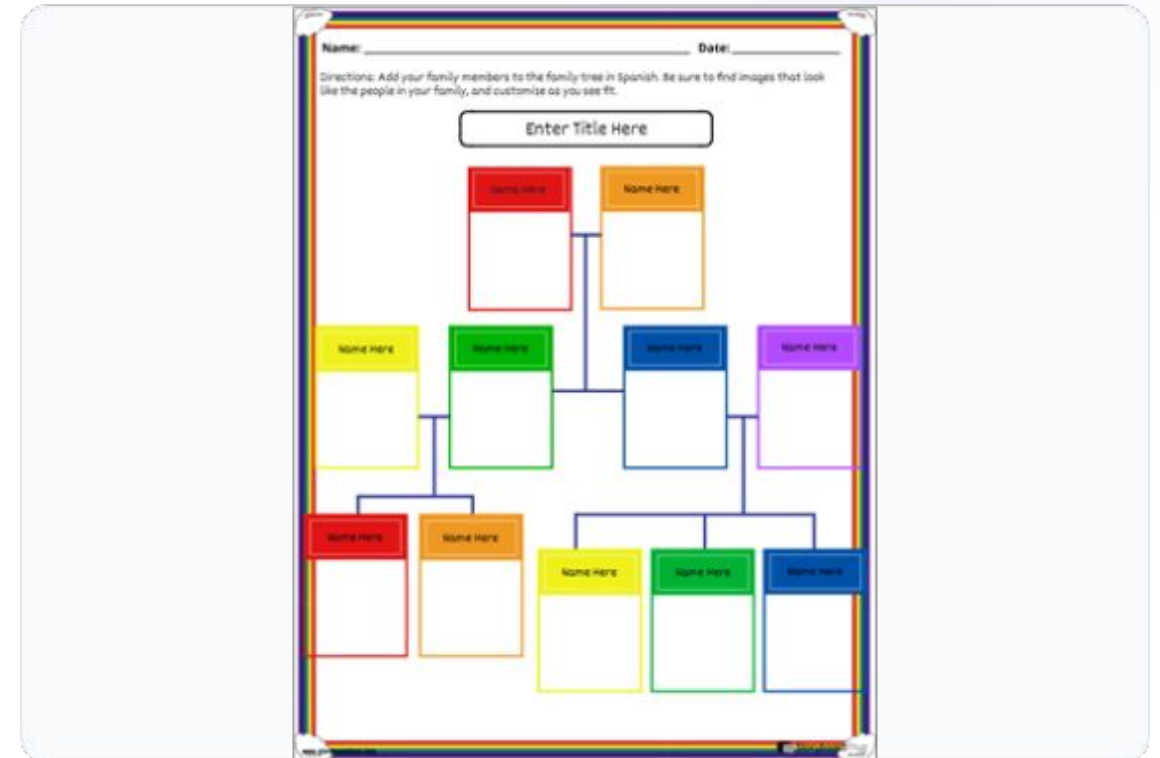
Generation 0 (Self): Brother, Sister, Cousin, Husband, Wife, In-laws.

Generation -1: Son, Daughter, Nephew, Niece, Son-in-law, Daughter-in-law.

Generation -2: Grandson, Granddaughter.

Family Tree Symbols

- ✓ **Square / Plus (+):** Used to represent male members clearly.
- ✓ **Circle / Minus (-):** Used to represent female members.
- ✓ **Double Horizontal Line (=):** Represents a married couple (Spouses).
- ✓ **Single Horizontal Line (-):** Connects siblings in the same generation.
- ✓ **Vertical Line (|):** Connects parents to their offspring across generations.



MODULE 02

Problem Solving Types

Mastering the three primary formats of questions asked in reasoning
examinations

Question Categories



1. Pointing / Introducing

A person introduces someone by describing a chain of relationships (e.g., "He is the son of the only daughter of my grandfather").



2. Family Tree Puzzles

Complex narrative snippets describing multiple family members and asking to find specific inter-relations between two people.



3. Coded Relations

Relations expressed using mathematical or special symbols (e.g., $A + B$ means A is father of B, $A * B$ means A is sister of B).

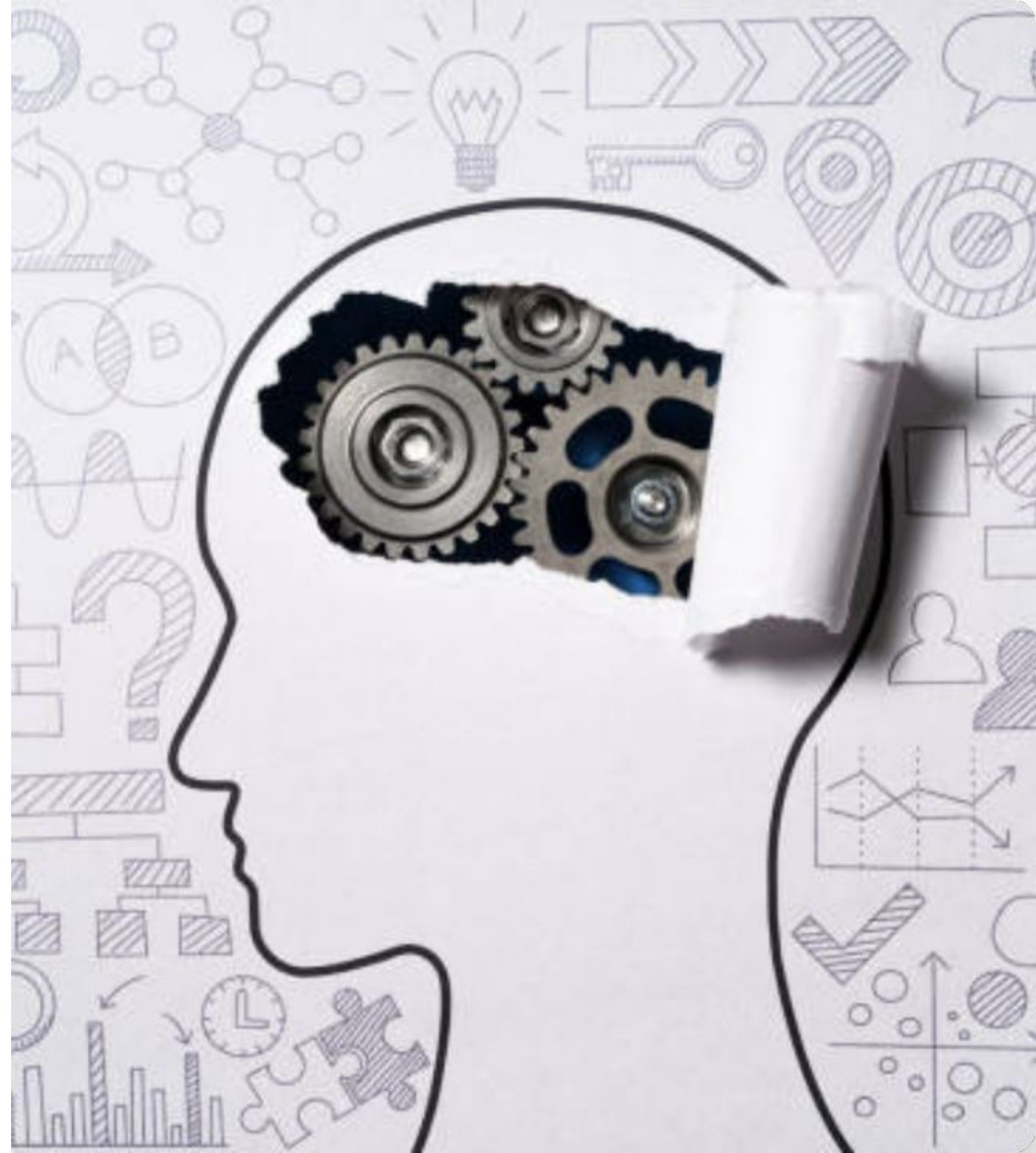
Pointing Type Strategy

The Backtracking Method

When tackling "pointing to a photo" questions, start from the word **"My"** or **"Me"** in the statement.

Break down the relationship step-by-step from right to left (backwards) to simplify the chain into a single direct relation.

Example: "My father's only daughter" = "My sister" or "Myself" (depending on speaker's gender).



Coded Relations Method

 **Gender Elimination:** Determine the required gender of the target person and instantly eliminate options where gender mismatches or is left unknown.

 **Generation Gap Method:** Assign numerical values to generational leaps (+1 for parent, 0 for sibling/spouse, -1 for child) to quickly evaluate expressions.

 **Right-to-Left Decoding:** Parse coded expressions like $P \$ Q \# R * S$ from right to left to construct family trees systematically.

Key Rules to Remember

0

Assumptions on
Gender

Golden Rule of Blood Relations

Never assume gender based on a person's name alone! Names like Robin, Kiran, or Sonu can belong to any gender in reasoning puzzles. Gender must strictly be deduced from explicit relational clues (e.g., "father", "sister", "wife", "son"). If gender cannot be determined from given conditions, choose *"Cannot be determined"*.

Relation Reference Table

Relationship Phrase	Direct Relation Term
Father's / Mother's Brother	Uncle (Paternal / Maternal)
Father's / Mother's Sister	Aunt (Paternal / Maternal)
Brother's / Sister's Son	Nephew
Brother's / Sister's Daughter	Niece
Uncle's / Aunt's Child	Cousin